Cost and QoS-Aware Service Migration in Vehicular Edge Computing Networks with Limited Backhaul Capacity

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***Abstract*—Vehicular edge computing (VEC) is a key enabler for providing low-latency vehicular service, by deploying the services at the edge servers associated with wireless nodes distributed in vehicle networks. Due to limited coverage of wireless nodes and high vehicle mobility, vehicular services need to be adaptively migrated over the backhaul network to follow the vehicles. Given an ever-increasing amount of vehicles served by finite backhaul bandwidth, the backhaul network can become the bottleneck for timely service migration. However, few studies on service migration considered optimizing the utilization of backhaul network bandwidth to alleviate such bottleneck. In this study, we focus on the service migration problem in VEC-enabled small cell networks with distributed wireless backhaul. Specifically, we formulate the service migration in VEC networks as an optimization problem with the objective of minimizing the migration cost while guaranteeing the desired quality of service (QoS), taking into account the limited time, bandwidth resource, and computation capacity. We develop an efficient service migration algorithm, called OMELET, which dynamically makes service migration decisions based on Lyapunov technique via jointly optimizing migration decision, path selection and rate allocation. Simulation results validate that OMELET can achieve a more effective trade-off between migration cost and QoS guarantee compared with the state-of-the-art service migration schemes.**

***Index Terms*—Vehicular edge computing, Distributed wireless backhaul, Service migration, Quality of service**

I. Introduction

Vehicular edge computing (VEC) has emerged as a promising paradigm to overcome the local vehicle resource bottleneck by deploying edge servers in the places near vehicles, such as cell networks [1]. With the continued expansion and innovation of fifth generation (5G) network, a huge number of small base stations (SBSs) are densely deployed to serve the exponentially growing data traffic and resort to a distributed wireless backhaul (DWB) communication mechanism to save the deployment cost and complexity [2], which greatly facilitates the service deployment in VEC. When a vehicle moves beyond the wireless coverage of its serving SBS, which is responsible for deploying and executing its service, the extended transmission distance increases communication latency, resulting in a significant decline in quality of service (QoS). Besides, the QoS can also be degraded when an SBS has insufficient resource capacity to process the service requests received from vehicles. Therefore, migrating services across different SBSs becomes crucial for maintaining QoS in the face of vehicle mobility and server resource limitations [3]. The integration of software-defined networking (SDN) and network functions virtualization within the VEC scenario enables network services and functions to be deployed and managed at the network edge as virtualized entities [4]. This integration significantly simplifies the orchestration of services, networks, and VEC resources, thus facilitating service migration across different SBSs to ensure service continuity.

Service migration consumes a certain amount of backhaul bandwidth to transfer service data, potentially leading to prohibitive migration costs. To tackle this challenge, research [5]–[7] has concentrated on the performance of service migration in VEC networks with emphasis on QoS and migration costs. However, the above studies assumed that services could be seamlessly migrated to any destination server without considering the impact of backhaul bandwidth on the timeliness of migration. In reality, migrating services encapsulated in virtual machines or containers is a time-consuming operation, especially under limited network bandwidth [8]. For instance, it has been demonstrated that migrating a running face recognition instance using live VM migration under 100 Mbps bandwidth takes over 300 seconds [9]. If the service is not fully migrated to the destination within the expected time, this migration will be deemed a failure, which merely wastes bandwidth without contributing to improving QoS. Therefore, service migration time, which is greatly influenced by bandwidth conditions, play a crucial role in determining the effectiveness and efficiency of the migration process.

In this paper, we explore an SDN-enhanced VEC network with densely situated SBSs connected by DWB. We formulate a cost-aware service migration problem that optimizes migration decision, path selection, and rate allocation to minimize the overall migration cost associated with transferring service data under the requirements of QoS guarantee. The primary contributions of this paper are outlined as follows.

- We present a DWB-interconnected SBS network architecture, where one SBS serves as the gateway, connecting to the core network via fiber-optic cables, while the remaining SBSs connect to the gateway through wireless backhaul. We investigate SDN-enhanced proactive ser-

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vice migration and develop a long-term migration cost minimization problem which jointly optimizes migration decision, path selection, and rate allocation under constraints of QoS, time and resource capacity.

- We propose an online service migration algorithm called OMELET, which exploits Lyapunov method to break down the temporally coupled problem into a sequence of one-time-slot lookahead sub-problems. To address these sub-problems, we present a priority-based algorithm which considers the service data size and workload for making migration decision and adopt a rate-orientated best-fit migration path selection strategy.
- We utilize real-world vehicle movement trajectories to perform extensive simulations where the proposed OMELET method is evaluated against three other algorithms including Least Hop [10], Cloud [6], and MoDEMS [11]. The results reveal that OMELET method can exploit the VEC network capacities effectively and achieves superior performance in reducing migration cost and ensuring QoS guarantee.

We organize the rest of the paper as follows. Section II summarizes the related work and their limitations. Section III presents the system model and formulates the optimization problem. The proposed method OMELET is elaborated in Section IV. The performance of OMELET is evaluated in Section V. Finally, the paper is concluded in Section VI.

II. Related Work

Edge computing provides mobile users with access to powerful resources within an intimate range, significantly reducing communication latency. Numerous previous studies have focused on exploring reactive service migration initiated by metric degradation, with user-orientated service relocation decisions made in response to detected QoS decline or mobility concerns involving edge nodes and users. Chen et al. [12] tackled the joint optimization of service migration and request routing in MEC environments by introducing a Lyapunov optimization-based online algorithm to enhance long-term network utility. Zhou et al. [13] developed an energy-efficient online algorithm that integrates Lyapunov and particle swarm optimizations to reduce average energy consumption while meeting service latency requirements and considering user interference. Xu et al. [14] proposed a novel game theory-inspired service migration strategy which models service revenue, delay, and cost into an auction framework to optimize the overall system benefit. Liu [15] presented a decision theory-based service migration method that analyzes vehicle mobility and optimizes service migration for service delay minimization and server workload balance. Peng et al. [16] developed a service migration scheme based on transfer reinforcement learning, aiming to maximize service satisfaction concerning delay by considering the joint impact of communication and computing costs.

Reactive service migration adjusts service placement only after performance declines, failing to preemptively address potential issues. To mitigate this limitation, proactive service migration techniques are explored to anticipate QoS metric deterioration and forecast the user mobility patterns for advance migration planning. Ding et al. [5] targeted energy and latency minimization, proposing service migration methods that cater to various mobility types. Badri et al. [17] designed a Sample Average Approximation-based algorithm to proactively manage resource demands and service migration, aiming to maximize system QoS within limited energy budget. Wang et al. [6] optimized microservice coordination to reduce service delay at low costs with a dynamic programming-based offline algorithm for global optimization and an online reinforcement learning algorithm for real-time adaptive strategies. Shah et al. [18] introduced an SDN-based migration technique with a unified control framework that manages service mobility by predicting application performance considering resource utilization, application characteristics, and QoS. Kim et al. [11] modeled the minimization of service provider costs and user latency under resource constraints and user mobility in VEC as an NP-hard integer linear programming problem, employing heuristic algorithms for solutions. Sfaxi et al. [7] developed a prediction method based on exponential smoothing for detecting QoS metric degradation in dynamic edge computing environments to guide the service migration. Wang et al. [19] designed a proactive migration strategy leveraging Lyapunov optimization and spatial-temporal mobility prediction for optimal migration decisions to ensure service reliability and reduce energy cost.

However, most existing studies have an assumption that the backhaul network has sufficient bandwidth capacity to support a large amount of service-orientated data to be migrated seamlessly in finite time period, which is inapplicable to the wireless backhaul with limited bandwidth. In this paper, we study the vehicular service migration optimization problem in the small cell network connected via distributed wireless backhaul. We construct a dynamic service migration framework that makes service migration decision based on the predicted vehicle mobility information considering migration time bound and resource constraints including server computation capacity and backhaul bandwidth.

III. System Model

We consider an SDN-assisted VEC platform in a DWB-enabled SBS network consisting of N vehicles denoted as $\mathcal{V} = \{1, 2, \ldots, N\}$ and S SBSs denoted as $\mathcal{S} = \{1, 2, \ldots, S\}$. One SBS acts as the gateway with fiber-optic connections to the core network and access to the cloud. Each SBS can communicate directly with SBSs within its coverage range through a one-hop wireless backhaul link. The gateway employs an SDN-based SBS controller to analyze vehicle mobility patterns, monitor server load, and manage network resources for effective service migration. We regard the Vehicle-Service-Edge computing system as a graph $G = (\mathcal{S}, \mathcal{E})$ where $\mathcal{E} = \{e_{ss'} | s, s' \in \mathcal{S}\}$ consists of all one-hop links between the SBSs in set $\mathcal{S}$. Each SBS is attached to an edge server which can provide computation services to vehicles. Consistent with [11], [19], we assume each vehicle $i \in \mathcal{V}$ is entitled to an

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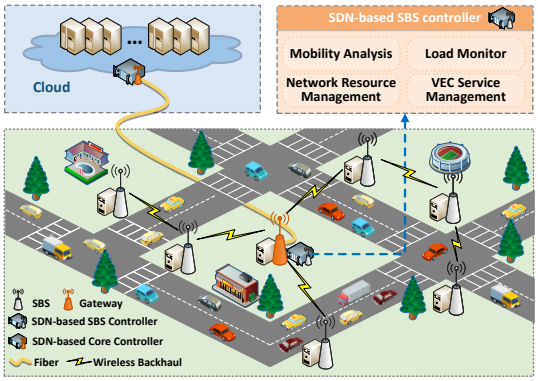

Fig. 1. The architecture of network

exclusive service instance whose running data size is denoted as D_i. For ease of capturing the mobility of vehicles, we consider a discrete time-slotted system $\mathcal{T} = \{0, 1, \ldots, T\}$ where each time slot $t \in \mathcal{T}$ has a length of χ seconds.

Each SBS communicates with vehicles exclusively within its designated coverage area through the radio access network. Let $l_{is}(t) \in \{0, 1\}$ denote whether vehicle i is associated with SBS s in time slot t where $l_{is}(t) = 1$ if associated and $l_{is}(t) = 0$ otherwise. For each time slot t, we define a binary service placement variable $x_{is}(t)$ where $x_{is}(t) = 1$ indicates SBS s deploys the service of vehicle i and functions as its serving SBS in time slot t and otherwise $x_{is}(t) = 0$. Following [11], [17], the service is placed at one and only one SBS at any time. Thus we have

$$\sum_{s \in \mathcal{S}} x_{is}(t) = 1. \tag{1}$$

To guarantee a low service latency and reliable service continuity, the service instance is proactively migrated based on the prediction of vehicle mobility. By leveraging advanced machine learning technologies, the future short-term mobility of vehicles can be precisely predicted [20]. Consistent with [5], we assume that in time slot t, the location variables of vehicles $\boldsymbol{l}(t+1) = \{l_{is}(t+1) | i \in \mathcal{V}, s \in \mathcal{S}\}$ in the subsequent time slot $t+1$ can be obtained through mobility prediction to guide proactive service migration one time slot ahead, ensuring accurate predictions for efficient migration.

A. QoS Model

In VEC environments, the QoS experienced by each vehicle is typically captured by service latency [5]. Without loss of generality, we assume that at the beginning of each time slot t, each vehicle i generates a service request (u_i, f_i), where u_i is the data volume produced during the interaction with the service, and f_i is the number of CPU cycles needed to fulfill the request. Each vehicle offloads its service requests to its associated SBS. If the associated SBS hosts the required service, it processes the requests locally; otherwise, it forwards them to the serving SBS via backhaul links. Due to limited backhaul bandwidth, forwarding requests between SBSs can significantly increase offloading latency. Thus, service latency is influenced by both the service location and the vehicle's position. Additionally, larger requests can further increase latency due to prolonged transmission process. Considering these factors, we define a QoS indicator for SBS s serving vehicle i in time slot t as follows:

$$QoS_{is}(t) = \frac{\gamma}{u_i(t) d_{ss'}(t)}, \tag{2}$$

where γ is a positive adjustment coefficient and $d_{ss'}(t)$ represents the distance between vehicle i's associated SBS s' ($l_{is'}(t) = 1$) and serving SBS s. The distance $d_{ss'}(t)$ between SBS s at coordinate (a_s, b_s) and SBS s' at coordinate $(a_{s'}, b_{s'})$ is computed using Euclidean distance function as $d_{ss'}(t) = \sqrt{(a_s - a_{s'})^2 + (b_s - b_{s'})^2}$. According to (2), the QoS experienced by vehicle i in time slot t is vulnerable to its request data volume and its distance to the serving SBS.

Given the service placement information, the QoS for vehicle i in time slot is:

$$QoS_i(t) = \sum_{s \in \mathcal{S}} x_{is}(t) QoS_{is}(t). \tag{3}$$

The QoS guarantees are often formalized through a Service Level Agreement (SLA), where each vehicle $i \in \mathcal{V}$ specifies a desired QoS level QoS_i^{exp}. An SLA violation occurs if the QoS for vehicle i falls below this threshold:

$$\phi_i(t) = \begin{cases} 0, & \text{if } QoS_i(t) \geq QoS_i^{\text{exp}}, \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

Additionally, to prevent overloading the SBS servers, we apply the following computation capacity constraints for each SBS:

$$\sum_{i \in \mathcal{V}} x_{is}(t) f_i \leq F_s, \tag{5}$$

where F_s is the maximum computation capacity of SBS j.

B. Service Migration Model

Each SBS is equipped with one antenna to receive signals and M antennas to send signals, where M is larger than the amount of SBSs within its communication coverage. We consider that SBSs conduct communication via a wireless backhaul based on an out-of-band full-duplex (OBFD) mode [21], where separate frequency bands are assigned to each SBS and each of its covering SBSs, enabling bidirectional communication between SBSs with no interference. We further utilize zero-forcing technology at each multi-antenna SBS to avoid the intra-cell interference [21]. Define W_s as the backhaul bandwidth allocated to SBS s. The achievable rate for the data transmission on link $e_{ss'}$ from SBS s to SBS s' can be calculated as

$$R_{e_{ss'}} = W_s \log_2 \left(1 + \frac{p_s}{||\boldsymbol{v}_{ss'}||^2 N_0}\right), \tag{6}$$

where p_s is the transmission power of SBS s, $\boldsymbol{v}_{ss'}$ is the zero-forcing beamforming vector obtained based on the channel state including path loss and fading factors and N_0 is the power of additive white Gaussian noise power.

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Define $\mathcal{P}_{ss'}$ to describe the set of available transfer paths from SBS s to SBS s', where each path $p_{ss'} \in \mathcal{P}_{ss'}$ can be denoted as a series of sequentially connected backhaul links $p_{ss'} = \{e_{ss_1}, e_{s_1 s_2}, \cdots, e_{s_k s'}\}$. Based on the transmission rates of individual links, we give the definition of transmission rate of path in Definition 1.

Definition 1 (Transmission Rate of Path)**.** *The transmission rate of path $p_{ss'}$, denoted as $R_{p_{ss'}}$, is defined as the minimum transmission rate among all backhaul links covered by $p_{ss'}$:*

$$R_{p_{ss'}} = \min_{e_{s_k s_{k'}} \in p_{ss'}} R_{e_{s_k s_{k'}}}. \tag{7}$$

Service migration requires the synchronization of vehicle-related data from the source SBS to the destination SBS to resume the service provisioning on the target SBS. To promptly transfer service data from the source SBS to the destination SBS, the appropriate migration path and migration rate should be determined. Denote a binary variable $o_{i,p_{ss'}}(t)$ to indicate whether path $p_{ss'} \in \mathcal{P}_{ss'}$ is chosen for migrating the service of vehicle i from SBS s to SBS s' during time slot t. Here, we assume that only one path is selected to transfer service data for each service migration. Hence, we have

$$x_{is}(t) + x_{is'}(t+1) - 1 \leq \sum_{p_{ss'} \in \mathcal{P}_{ss'}} o_{i,p_{ss'}}(t) \leq x_{is}(t). \tag{8}$$

The service migration taking place in the backhaul network includes transmission, propagation, processing, and queuing. Since the delay of propagation, processing, and queuing mainly depends on the hop count between the two SBSs, the migration time for vehicle i to transfer its service data from SBS s to SBS s' via path $p_{ss'}$ can be computed as follows:

$$\tau^m_{i,p_{ss'}}(t) = \begin{cases} \frac{D_i}{r_{i,p_{ss'}}(t)} + \lambda h(p_{ss'}), & s \neq s', \\ 0, & s = s', \end{cases} \tag{9}$$

where $r_{i,p_{ss'}}(t)$ is the transmission rate of path $p_{ss'}$ assigned to vehicle i for service migration, $h(p_{ss'})$ is the hop count along path $p_{ss'}$ between SBS s and SBS s', and λ is a positive coefficient. Denoting the set of paths covering link $e_{s_k s_{k'}}$ as $\mathcal{P}(e_{s_k s_{k'}})$, we have the following maximum link rate constraints for each link imposed by (6):

$$\sum_{i \in \mathcal{V}} \sum_{p_{ss'} \in \mathcal{P}(e_{s_k s_{k'}})} o_{i,p_{ss'}}(t) r_{i,p_{ss'}}(t) \leq R_{e_{s_k s_{k'}}}. \tag{10}$$

Accordingly, the service migration time can be computed as

$$T^m_i(t) = \sum_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} \sum_{p_{ss'} \in \mathcal{P}_{ss'}} o_{i,p_{ss'}}(t) \tau^m_{i,p_{ss'}}(t). \tag{11}$$

Ensuring QoS hinges on the migration time, which must not surpass the time limit required for seamless service migration. Recall that we consider one-time-slot lookahead service migration given the decrease of mobility prediction accuracy over time. In such case, the service migration should be finished before the beginning of the next time slot to provide a timely and stable service response for the newly arrived vehicles' requests. Thus, we have:

$$T^m_i(t) \leq \chi. \tag{12}$$

Data transferring during service migration consumes a lot resources, leading to considerable migration overheads. As noted in [12], the migration cost primarily depends on the size of service data to be transmitted. Accordingly, we define the migration cost incurred from proactively migrating the service of vehicle i as follows:

$$c^m_i = cD_i, \tag{13}$$

where c is the cost of transmitting 1 Mb of data over the wireless backhaul. Subsequently, the migration cost for vehicle i in time slot t can be calculated as

$$C^m_i(t) = \sum_{s \in \mathcal{S}} \sum_{s' \in \mathcal{S}} \sum_{p_{ss'} \in \mathcal{P}_{ss'}} o_{i,p_{ss'}}(t) c^m_i. \tag{14}$$

C. Problem Formulation

According to (4), we can obtain the average SLA violation rate across all vehicles in time slot t as

$$\bar{\phi}(t) = \frac{1}{N} \sum_{i \in \mathcal{V}} \phi_i(t). \tag{15}$$

To maintain QoS, the average long-term SLA violation rate over multiple time slots must remain below a specified threshold $\Phi \in (0, 1]$. Our goal is to perform service migration at the lowest possible cost to provide the desired long-term SLA violation rate. Considering optimizing service placement $\boldsymbol{x} = \{x_{is}(t), \forall i \in \mathcal{V}, s \in \mathcal{S}, t \in \mathcal{T}\}$, migration path selection $\boldsymbol{o} = \{o_{i,p_{ss'}}(t), \forall i \in \mathcal{V}, p_{ss'} \in \mathcal{P}, t \in \mathcal{T}\}$, and migration rate allocation $\boldsymbol{r} = \{r_{i,e_{ss'}}(t), \forall i \in \mathcal{V}, e_{ss'} \in \mathcal{E}, t \in \mathcal{T}\}$, we build the long-term total migration cost minimization problem:

$$\min_{\boldsymbol{x},\boldsymbol{o},\boldsymbol{r}} \sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{V}} C^m_i(t) \tag{16a}$$

$$\text{s.t.} \lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T} \bar{\phi}(t) \leq \Phi, \tag{16b}$$

$$x_{is}(t), o_{i,p_{ss'}}(t) \in \{0, 1\}, \tag{16c}$$

$$(1), (5), (8), (10), (12).$$

IV. Algorithm Design

A. Problem Transformation

A key challenge in solving Problem (16) is the requirement for complete foresight into vehicle mobility, as specified by Constraint (16b). However, as noted by [20], predictions of vehicle movements based on machine learning tend to lose accuracy over longer periods, making precise long-term forecasting impractical. To overcome this limitation, we employ Lyapunov optimization with a one-time-slot lookahead prediction. By redefining SLA compliance as a queue stability problem, the method uses short-term movement prediction to enable precise and proactive service migration without relying on long-term forecasts. This is achieved through a virtual queue mechanism defined as:

$$O(t+1) = \max[O(t) + \bar{\phi}(t+1) - \Phi, 0], \tag{17}$$

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where $O(t)$ tracks the accumulation of SLA violation rate beyond threshold Φ in the end of time slot t with an initial value of $O(0) = 0$. According to [22], stabilizing $O(t)$ ensures adherence to the SLA violation constraint, as given by:

$$\lim_{t\to\infty} \frac{\mathbb{E}[O(t)]}{t} = 0. \tag{18}$$

We employ the drift-plus-penalty approach [22] to convert Problem (16) to a set of slot-level problems. We begin by defining the quadratic Lyapunov function as:

$$L\left(O(t)\right) \triangleq \frac{1}{2}O(t)^2. \tag{19}$$

The one-step Lyapunov drift function is then defined as:

$$\Delta O(t) \triangleq \mathbb{E}[L\left(O\left(t+1\right)\right) - L(O(t))|O(t)], \tag{20}$$

which quantifies the variation of Lyapunov function per time slot. The minimization of $\Delta O(t)$ is crucial for maintaining the stability of virtual queue $O(t)$. To merge the objectives of minimizing Lyapunov drift and service migration cost, we formulate the Lyapunov drift-plus-penalty function as follows:

$$\Delta(t) = \Delta O(t) + V\mathbb{E}[C^m(t)|O(t)], \tag{21}$$

where $C^m(t) = \sum_{i\in\mathcal{V}} C_i^m(t)$, and V is a non-negative weight factor that modulates the significance of the SLA violation rate queue backlog relative to the service migration cost.

Theorem 1. *For any value of $O(t)$, the drift-plus-penalty function can be upper-bounded as*

$$\begin{aligned}\Delta O(t) + V\mathbb{E}[C^m(t)|O(t)] &\le \frac{1}{2} \\ &+ O(t)\mathbb{E}[\bar{\phi}(t+1) - \Phi|O(t)\} + V\mathbb{E}[C^m(t)|O(t)].\end{aligned} \tag{22}$$

Proof. It is obvious that $(\max[i,0])^2 \le i^2$ for any $i \in \mathbb{R}$. Therefore, we have:

$$\begin{aligned}&L(O(t+1) - L\left(O(t)\right) && \text{(23a)}\\ &= \frac{1}{2}\left[\left(\max\left[O(t) + \bar{\phi}(t+1) - \Phi, 0\right]\right)^2 - O^2(t)\right] && \text{(23b)}\\ &\le \frac{1}{2}\left[(O(t) + \bar{\phi}(t+1) - \Phi)^2 - O^2(t)\right] && \text{(23c)}\\ &\le \frac{1}{2}(\bar{\phi}(t+1) - \Phi)^2 + O(t)(\bar{\phi}(t+1) - \Phi). && \text{(23d)}\end{aligned}$$

Note that $\bar{\phi}(t+1) \in [0,1], \forall t \in \mathcal{T}$ and $\Phi \in (0,1]$, which infer $(\bar{\phi}(t+1) - \Phi)^2 \le 1, \forall t \in \mathcal{T}$. Hence, we have

$$L(O(t+1) - L\left(O(t)\right) \le \frac{1}{2} + O(t)(\bar{\phi}(t+1) - \Phi). \tag{24}$$

Combining these insights with the drift-plus-penalty framework results in (22). □

Following the method of opportunistically minimizing an expectation as described in [22], we minimize objective function (16a) through the greedy minimization of the right-hand side of inequality (22) given observed value of $O(t)$ in each time slot. By excluding the constant term $1/2$, Problem (16) simplifies to the following one-time-slot lookahead subproblems:

$$\min_{\boldsymbol{x}(t+1),\boldsymbol{o}(t),\boldsymbol{r}(t)} \quad \Theta(t) = O(t)(\bar{\phi}(t+1) - \Phi) + VC^m(t), \tag{25a}$$

$$\text{s.t.} \quad (1), (5), (8), (10), (12), (16c).$$

Nevertheless, Problem (25) remains non-convex, with its intricate objective function and discrete constraints complicating the solution process. To address Problem (25), we introduce a greedy method aimed at obtaining a suboptimal solution.

B. Online Service Migration for VehicuLar Edge Computing

At the beginning of time slot t, we leverage the predicted location of vehicles, $\boldsymbol{l}(t+1)$, of the upcoming time slot $t+1$ to initiate proactive service migrations. Based on predictive information about vehicle positions in the next time slot, the services to be migrated and their migration destinations are screened on the premise of QoS improvement. Specifically, a vehicle i belonging to the migration candidate set $\mathcal{V}^m(t) = \{i \in \mathcal{V} | \sum_{s\in\mathcal{S}} x_{is}(t) QoS_{is}(t+1) < QoS_i^{\text{exp}}\}$ can be allowed to migrate its service. This screening criterion ensures that only vehicles anticipated to violate the SLA constraints in the upcoming time slot are considered for proactive migration. Then, for vehicle $i \in \mathcal{V}^m(t)$, a destination SBS s' is chosen from the available destination set $\mathcal{S}_i(t+1) = \{s \in \mathcal{S} | QoS_{is}(t+1) \ge QoS_i^{\text{exp}}\}$, ensuring that the service can migrate only to SBSs that are expected to meet the required QoS level in the subsequent time slot. By implementing these two screening criteria, the efficacy of the migration in improving QoS is maintained, and the solution space is effectively reduced.

Then we propose a priority-based service migration decision mechanism which considers the service data size and workload and a rate-orientated best-fit migration path selection strategy as illustrated in Algorithm 1. An effective migration strategy involves migrating services that help control SLA violation rates while incurring minimal migration costs, aligning with the optimization objective function (25a). The above two screening criteria ensures that each migration instance contributes equally to reducing the future SLA violation rate. Therefore, priority should be given to migrating services with lower migration costs. According to (13), the service migration cost is proportional to the service data size D_i. Additionally, considering the computational constraint at each SBS, migrating services with lighter workloads can improve the success rate of migrations. Thus, we define a migration priority indicator for each vehicle $i \in \mathcal{V}^m(t)$ as follows:

$$P_i = \frac{\epsilon}{D_i f_i}, \tag{26}$$

where ϵ is a weight parameter. Accordingly, services with smaller D_i and f_i are given higher priority for migration.

To preserve SBS computation and backhaul network resources for subsequent migrations of computation-intensive and large-size services, a best-fit joint destination SBS and path selection approach is employed. For the service of vehicle

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$i \in \mathcal{V}^m(t)$, the SBSs in available destination set $\mathcal{S}_i(t+1)$ are arranged in ascending order of remaining computation resource $\tilde{F}_j$. The SBS in $\mathcal{S}_i(t+1)$ with the lowest $\tilde{F}_{s'}$ that can handle the service computation request ($f_i \leq \tilde{F}_{s'}$) is selected as the destination. Then available transmission paths $p_{ss'}$ from the service's source SBS s to SBS s' are arranged in ascending order of remaining transmission rate $\tilde{R}_{p_{ss'}}$. The minimal transmission rate for migrating vehicle i's service is determined as follows:

Proposition 1. *The minimal transmission rate required to complete the service migration when selecting path $p_{ss'}$ for transferring the service data of vehicle i is given by*

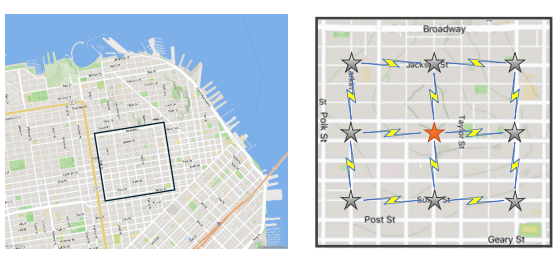
$$r^{min}_{i,p_{ss'}} = \frac{D_i}{\chi - \lambda h(p_{ss'})}, \tag{27}$$

according to time constraint (12).

The path with the lowest $\tilde{R}_{pss'}$ that can handle the migration request ($r^{\min}_{i,p_{ss'}} \leq \tilde{R}_{p_{ss'}}$) is selected to transfer the service data of vehicle i from source SBS s to destination SBS s' at a rate of $r^{\min}_{i,p_{ss'}}$. After selecting the path, the available transmission rates of the backhaul links on this path and the computation resource of source SBS and destination SBS are updated. For each solution update, the new objective value is obtained based on (25a), varying from $\tilde{\Theta}_\xi$ to $\tilde{\Theta}_{\xi^*}$. The iteration ends if the objective value is no longer reduced.

V. Performance Evaluation

In this section, we conduct simulations to evaluate the performance of our proposed OMELET method in terms of migration cost and QoS assurance. Our simulations are based on real-world cab mobility data collected in San Francisco [23]. Specifically, we select a 1.2 km $\times$ 1.2 km area from San Francisco, as indicated by the rectangular region in Fig. 2(a), and filter the mobility traces within this area for our simulation. The network topology adopted in the simulation is illustrated in Fig. 2(b), where nine SBSs are deployed to provide services for vehicles within a coverage radius of 300 m with eight SBSs marked with silver stars and one gateway SBS with an orange star. We normalize the beamforming vector $||\boldsymbol{v}||$ and noise power N_0 to 1. The parameters are primarily set according to [6], and are listed in Table 1, where the parameters taking values in an interval follow a uniform distribution in this interval.



(a) Simulation area selected from San Francisco (b) The network topology of simulation

Fig. 2. Simulation senario

Algorithm 1 Online service Migration for vEhicuLar Edge compuTing (OMELET)

Input: $\boldsymbol{x}(t)$
Output: $\boldsymbol{\xi}(t) = \{\boldsymbol{x}(t+1), \boldsymbol{o}(t), \boldsymbol{r}(t)\}$

```
Aquire the predicted vehicle location l(t+1);
Initialize x(t+1) ← x(t), o(t) ← 0, r(t) ← 0;
Sort V^m(t) in the descending order of P_i;
for i ∈ V^m(t) do
    Calculate Θ_ξ under current solution ξ(t);
    Sort S_i(t+1) in the ascending order of F̃_s;
    for s' ∈ S_i(t+1) do
        if f_i ≤ F̃_s' then
            Obtain the available transmission path set P_ss';
            Sort P_ss' in the ascending order of R̃_p_ss';
            for p_ss' ∈ P_ss' do
                Calculate the required minimal transmission rate r^min_i,p_ss' of p_ss' according to (27);
                if r^min_i,p_ss' ≤ R̃_p_ss' then
                    Assign x_is'(t+1) ← 1, o_i,p_ss'(t) ← 1, {r_i,e_sk s_k'(t) ← r^min_i,p_ss', ∀e_sk s_k' ∈ p_ss'};
                    Update F̃_s ← F̃_s + f_i, F̃_s' ← F̃_s' − f_i, {R̃_e_sk s_k' ← R̃_e_sk s_k' − r^min_i,p_ss', ∀e_sk s_k' ∈ p_ss'};
                    goto continue-outer-loop;
                end if
            end for
        end if
    end for
    continue-outer-loop:
    Calculate Θ_ξ* under updated solution ξ*(t);
    if Θ_ξ* > Θ_ξ then
        End the iteration and return ξ(t);
    end if
end for
```

A. Performance and Convergence

We evaluate the impact of V on OMELET's average migration cost and average SLA violation rate across time slots and queue length variation, as shown in Fig. 3. In Fig. 3(a), the average migration cost descends with V while the average SLA violation rate keeps growing. This is because the increase of V shifts OMELET's optimization objective towards a greater focus on reducing migration cost during the service migration decision-making process by adding the weight of migration cost. Inspired by this, on the premise of SLA violation rate guarantee, the value of parameter V should be set as large as possible to save migration cost. According to Fig. 3(a), we select $V = 0.09$ as the optimal values of V for our design, where the migration cost reaches the lowest point while the SLA violation rate is controlled under the desired level $\Phi = 0.1$. The increase of SLA violation rate with V leads to a rising queue length of OMELET with V, as shown in Fig. 3(b). Despite the queue length volatility resulting from the shifty value of V, the queuing system state can always be stabilized since the stability constraint can be guaranteed.

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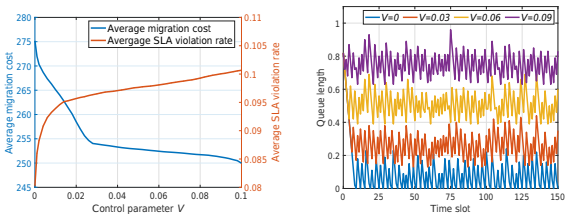
TABLE I
SIMULATION PARAMETERS

Parameter	Value
Number of vehicles N	200
Number of SBSs S	9
Service request size u_i	200 Mbits
Service computation requirements f_i	1 Ghz
Service data size D_i	$[400, 800]$ Mbits
SBS computation capacity F_s	25 Ghz
Transmission power of SBSs P_s	4W
SBS backhaul bandwidth W_s	$[30, 60]$ MHz
QoS requirement	$[0.45, 0.9]$
QoS adjustment coefficient γ	3.6×10^4
Hop latency coefficient λ	0.01 s
Migration cost c	0.01 per Mbit
SLA violation rate threshold Φ	0.1
Time interval χ	30 seconds



(a) Migration cost and SLA violation rate

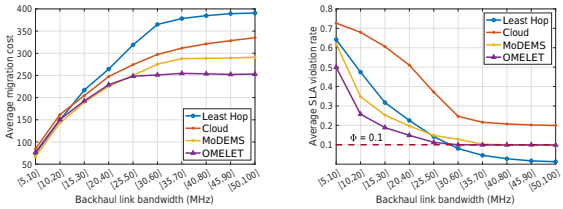
(b) Queue length

Fig. 3. Impact of V on the average migration cost, average SLA violation rate and queue length

B. Method Comparison

We compare OMELET with three other methods, including Least Hop [10], Cloud [6], and MoDEMS [11]. Least Hop [10] proactively migrates the services to the SBS closest to the vehicle's location in next time slot aiming at minimizing the communication latency between vehicle and services. Cloud [6] studies the online service migration based on predicted information to minimize user costs which however neglects the resource constraints. MoDEMS [11] is a heuristic service migration technique which minimizes the resource cost and user dissatisfaction cost based on mobility prediction without considering migration path selection.

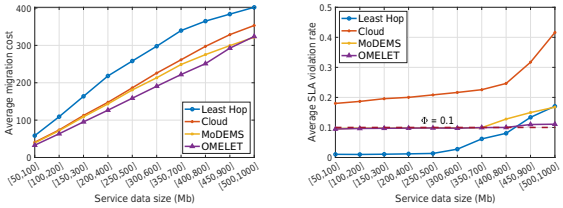
The impact of backhaul link bandwidth: As observed from 4(a) and 4(b), with the increase of backhaul bandwidth, the



(a) Average migration cost.

(b) Average SLA violation rate.

Fig. 4. The impact of backhaul link bandwidth on average the migration cost and SLA violation rate considering different methods.



(a) Average migration cost.

(b) Average SLA violation rate.

Fig. 5. The impact of service data size on average the migration cost and SLA violation rate considering different methods.

migration costs grow while the SLA violation rates descends. This is because the limited backhaul bandwidth can be a bottleneck that limits the amount of service migration and the growth of backhaul bandwidth allows more services to be migrated in time. Under the low-bandwidth backhaul where the bandwidth is in a range lower than $[30, 60]$ MHz, OMELET achieves a lower SLA violation rate than the other methods at a migration cost near the lowest level. This is because OMELET can fully utilize the limited bandwidth resource to support more prompt service migration by making priority-based service migration decisions considering migration candidate screening and migration path planning. When the backhaul bandwidth rises to the range of $[30, 60]$ MHz, OMELET manages to keep the SLA violation rate under the required level at the minimal migration cost among all the methods. These observations demonstrate that OMELET is able to achieve more efficient service migration with varying network bandwidth conditions compared with the other methods.

The impact of service data size: Since all service data need to be migrated for seamless service execution, the migration cost increases as the size of service data rises as shown in Fig. 5(a). Furthermore, the increase of service data transmission demands higher backhaul bandwidth to meet both effectiveness and timeliness requirements of the service migration. Thus, the SLA violation rate rises as the available backhaul network bandwidth gets far from being enough for meeting the service data migration demands as seen in Fig. 5(b). In Fig. 5(b), Cloud achieves the highest increase of SLA violation rate due to its inefficient migration planning which neglects the capacity of backhaul network and SBS server. Throughout the whole size increase process, OMELET is able to provide an SLA violation rate closest to the threshold at a steadily lowest migration cost among all the methods. This is because that OMELET considers the service data size and backhaul capacity to flexibly adjust the migration plan to minimize the migration cost while following the QoS constraint.

The impact of number of vehicles: The growing number of vehicles indicates an increase of the services need to be placed and migrated which intensifies the resource competition, leading to a rising migration cost and SLA violation rate in Fig. 6. The limited backhaul capacity imposes a ceiling on the transmission load. Thus the migration costs of the four methods positively associated with the migration load nearly

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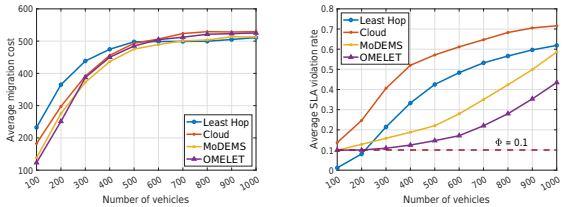

(a) Average migration cost. (b) Average SLA violation rate.

Fig. 6. The impact of number of vehicles on average the migration cost and SLA violation rate considering different methods.

stop increasing and stay at a comparable level as the bandwidth resource becomes depleted by the growing migration load as shown in Fig. 6(a). In Fig. 6(b), OMELET gains a lower SLA violation rate than Least Hop as the number of vehicles increases beyond 200. This is because OMELET can fully utilize the time and resources to enable as many services as possible to be migrated to their optimal SBSs according to the vehicle movement under the resource stress. Without the screening of migration candidates and destinations, Least Hop simply migrates each vehicle's service to the nearest SBS based on the predicted vehicle position. This intuitive migration strategy results in a high migration cost and consumes a large amount of bandwidth which can degrade the perceived QoS especially when the transferred volume becomes large. In summary, OMELET achieves the greatest comprehensive performances including migration cost-saving and QoS guarantee.

VI. Conclusion

In this paper, we study the dynamic VEC service migration problem in SDN-enabled cellular network interconnected by DWB. This problem aims to minimize the service migration cost while guaranteeing the desired QoS level considering the time and resource constraints. We propose a priority-based algorithm to perform the service migration with prior knowledge of next-time-slot vehicle mobility. Extensive simulations have been conducted to evaluate the effectiveness and efficiency of the proposed algorithm. For future work, the joint optimization of migration decision and resource allocation for long-term service migration which can span time slots by introducing the mobility prediction accuracy is a practical and urgent problem to be investigated.

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